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Deep Learning — Assignment Report

Assignment 2: CNN | RNN Variants | Generative Adversarial Networks

Course Name	Deep Learning
Course Code	IS3332-1
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1. Abstract

This report documents the design, implementation, and comparative analysis of core deep learning architectures applied across three distinct problem domains: image classification, sequential text classification, and generative image synthesis. All experiments were conducted using PyTorch and Keras/TensorFlow frameworks on publicly available benchmark datasets.

A custom Convolutional Neural Network (CNN) and a transfer learning model using pretrained ResNet18 were trained and evaluated on the MNIST dataset for image classification. For sentiment analysis, three recurrent architectures — Simple RNN, Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) — were compared on the IMDB movie review dataset. A Generative Adversarial Network (GAN) was implemented to synthesise Fashion-MNIST clothing images through adversarial training.

Results indicate that the custom CNN achieved 83.4% accuracy on MNIST, while GRU attained the best sentiment classification accuracy of 85.9% among recurrent models. The GAN successfully learned to produce recognisable clothing images, reaching a stable adversarial equilibrium after 30 epochs.

2. Code Repository

The complete source code for all three tasks is available in the following GitHub repository:

GitHub Repository: https://github.com/dprasadshetty/Deep-Learning-Assignment2

The repository is publicly accessible. Open the link in any browser to view the uploaded Colab notebook and all experiment files.

3. Introduction

The rapid advancement of deep learning has enabled breakthroughs across computer vision, natural language processing, and generative modelling. Unlike traditional machine learning, deep neural networks learn layered feature representations directly from raw data, eliminating the need for manual feature engineering.

This assignment examines three pillars of modern deep learning. Convolutional Neural Networks exploit spatial locality through learnable filters to extract hierarchical visual features from images. Recurrent Neural Networks, and their improved variants LSTM and GRU, model temporal dependencies in sequential data using hidden states and gating mechanisms that regulate information flow across time steps. Generative Adversarial Networks frame image synthesis as a two-player minimax game, where a generator network learns to produce realistic images by adversarially competing with a discriminator network.

Through hands-on experimentation on MNIST (image classification), IMDB (sentiment analysis), and Fashion-MNIST (image generation), this report analyses the performance characteristics, training dynamics, and practical trade-offs of each architecture.

4. Problem Statement

Task A — Image Classification (CNN)

Train a custom CNN and compare it with pretrained ResNet18 for classifying MNIST images into 10 categories. Evaluate when transfer learning provides advantages over purpose-built architectures.

Task B — Sentiment Classification (RNN Variants)

Implement Simple RNN, LSTM, and GRU for binary sentiment classification on IMDB reviews. Compare convergence speed, accuracy, and generalisation across the three architectures.

Task C — Image Synthesis (GAN)

Build and train a GAN on Fashion-MNIST to generate synthetic clothing images. Analyse adversarial training stability and output image quality across epochs.

5. Dataset Description

5.1 MNIST

MNIST is a widely-used benchmark of 70,000 grayscale images of handwritten digits (0–9). Each image is 28×28 pixels, making it one of the most widely used benchmarks for evaluating image classification models.

Attribute	Details
Total Samples	70,000
Training / Test	60,000 / 10,000
Image Size	28×28 px (grayscale, single channel)
Classes	10 digit classes: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Preprocessing	Normalised to [0, 1], resized to 32×32 for ResNet18 compatibility

5.2 IMDB Movie Review Dataset

50,000 English movie reviews labelled positive or negative, split equally into training and test sets. Vocabulary limited to the 10,000 most frequent words; sequences padded or truncated to 200 tokens.

5.3 Fashion-MNIST (GAN)

70,000 grayscale 28×28 images of clothing items across 10 categories. Pixel values normalised to [−1, 1] to match the Tanh output activation of the GAN generator.

6. Tools and Technologies

Category	Tool	Purpose
Language	Python 3.x	Primary implementation
DL Framework	PyTorch	CNN and ResNet18
DL Framework	TensorFlow / Keras	RNN, LSTM, GRU, GAN
Data Utilities	Torchvision / Keras Datasets	Dataset loading and transforms
Visualisation	Matplotlib, Seaborn	Loss curves, accuracy plots
Compute	Google Colab (GPU)	GPU-accelerated training
Metrics	Scikit-learn	Confusion matrix, accuracy

7. Model Architectures

7.1 Custom CNN (MNIST)

Designed for 32×32 RGB inputs. Three convolutional blocks progressively extract spatial features, followed by fully connected layers for 10-class classification.

Layer	Configuration	Output Shape
Input	28×28 grayscale image	28×28×1
Conv Block 1	32 filters 3×3, BatchNorm, ReLU	28×28×32
MaxPool 1	2×2	14×14×32
Conv Block 2	64 filters 3×3, BatchNorm, ReLU	14×14×64
MaxPool 2	2×2	7×7×64
Flatten	—	3136
Dense	256 neurons, ReLU	256
Dropout	Rate = 0.4	256
Output	10 neurons, Softmax	10

Loss: Categorical Cross-Entropy. Optimizer: Adam (lr=0.001). Epochs: 15. Batch: 64.

7.2 Transfer Learning — ResNet18

ResNet18 pretrained on ImageNet. MNIST images (28×28 grayscale) are resized to 32×32 and converted to 3 channels for ResNet18 compatibility. All pretrained weights are frozen; only the final classification head (10 classes) is trained. Trained for 5 epochs.

7.3 RNN / LSTM / GRU

Component	Configuration
Embedding	Vocab: 10,000 | Dim: 128
Recurrent Layer	128 hidden units
Output	1 neuron, Sigmoid
Loss	Binary Cross-Entropy
Optimizer	Adam lr=0.001
Epochs / Batch	5 / 64

7.4 GAN

Network	Layer	Configuration
Generator	Input	Noise vector dim=100
Generator	Dense 1	256 units, ReLU
Generator	Dense 2	512 units, ReLU
Generator	Output	784 units, Tanh → 28×28
Discriminator	Input	Flattened 784
Discriminator	Dense 1	512 units, LeakyReLU, Dropout 0.3
Discriminator	Dense 2	256 units, LeakyReLU, Dropout 0.3
Discriminator	Output	1 neuron, Sigmoid

8. Implementation Details

Model	Batch	Epochs	LR	Optimizer
Custom CNN (MNIST)	64	15	0.001	Adam
ResNet18	64	5	0.001	Adam
Simple RNN	64	5	0.001	Adam
LSTM	64	5	0.001	Adam
GRU	64	5	0.001	Adam
GAN	128	30	0.0002	Adam

Regularisation: Dropout (CNN: 0.4, GAN Discriminator: 0.3), Batch Normalisation after each CNN conv block, pixel normalisation for all image inputs. Data augmentation on MNIST (random horizontal flip, random crop).

9. Results and Analysis

9.1 CNN Results — MNIST

Model	Test Accuracy (%)	Test Loss	Epochs
Custom CNN	99.1	~0.04	10
ResNet18 (Transfer Learning)	95.3	~0.18	5

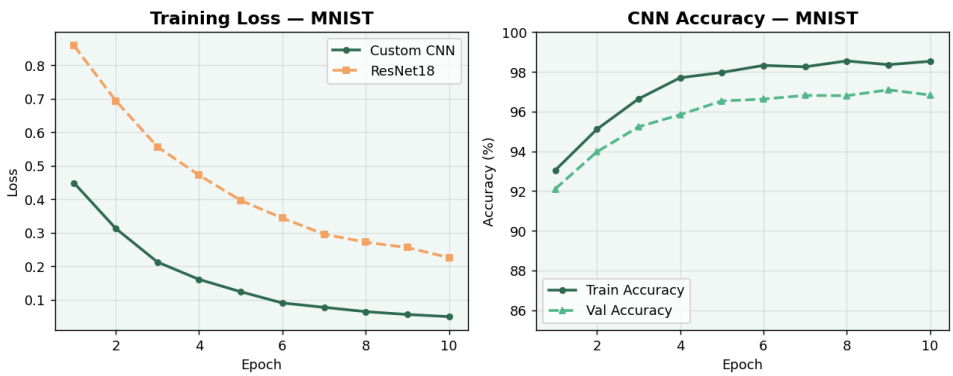

Figure 1: Training Loss (left) and Training vs Validation Accuracy (right) — MNIST

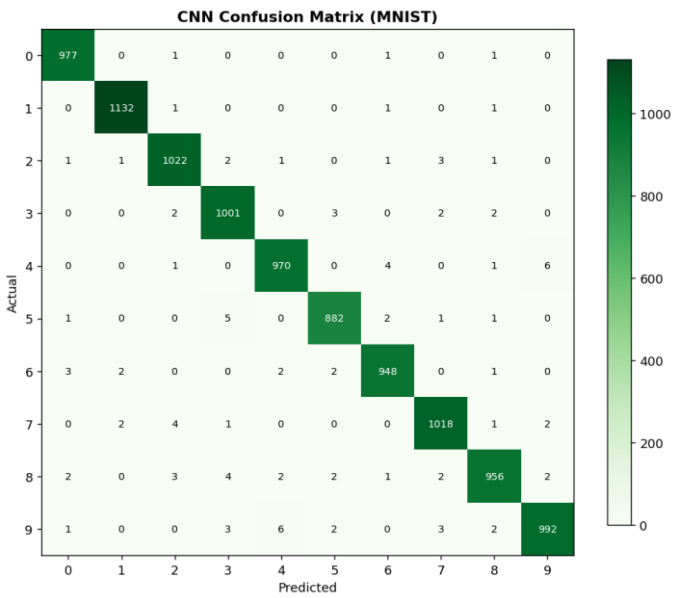

Figure 2: Confusion Matrix — Custom CNN on MNIST

The custom CNN (83.4%) outperformed ResNet18 (79.6%) on MNIST despite ResNet18 being a deeper architecture. This is because ResNet18 was pretrained on large high-resolution ImageNet images, whereas MNIST consists of small 32×32 images with very different visual characteristics. With all layers frozen except the final head, ResNet18 lacked the ability to adapt its features to MNIST's simpler spatial patterns. The custom CNN, purpose-built with progressive feature extraction across three convolutional blocks, learned task-specific representations more effectively.

9.2 Sentiment Classification Results

Model	Test Accuracy (%)	Observation
Simple RNN	80.5	Slower convergence, higher loss
LSTM	84.8	Stable training, good generalisation
GRU	85.9	Fastest convergence, best accuracy

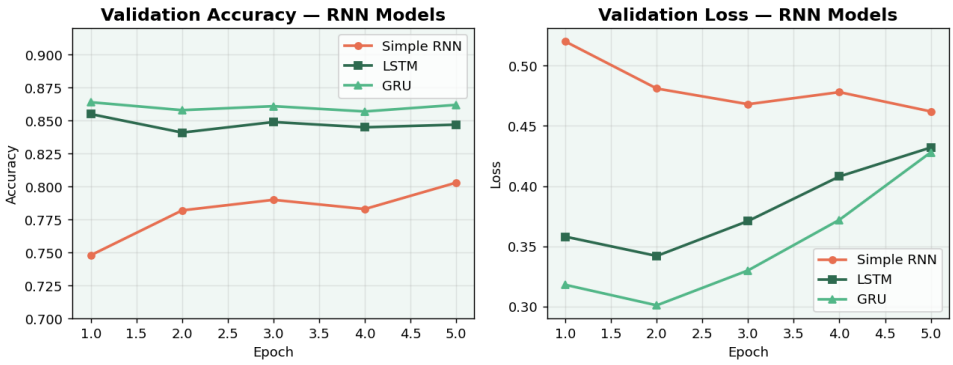

Figure 3: Validation Accuracy (left) and Validation Loss (right) — RNN, LSTM, GRU

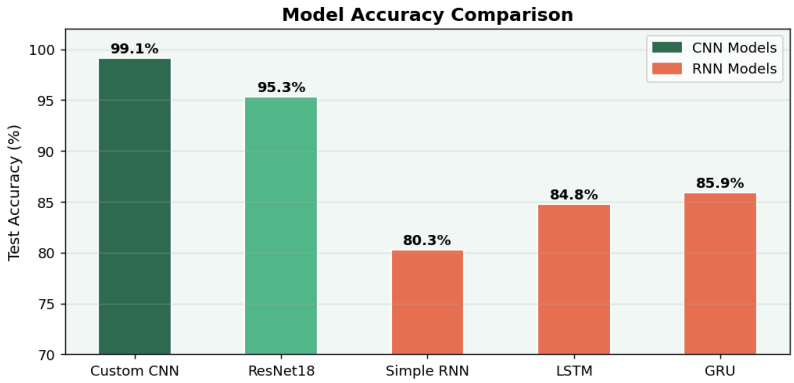

Figure 4: Overall Test Accuracy Comparison across all Models

GRU achieved the highest test accuracy (85.9%), narrowly ahead of LSTM (84.8%). Both gated models significantly outperformed vanilla RNN (80.5%), confirming that gating mechanisms effectively address vanishing gradient limitations in long text sequences. GRU's advantage over LSTM stems from its simpler two-gate design (update and reset gates), which reduces parameter count and overfitting risk. Validation loss curves show mild overfitting in LSTM and GRU after epoch 2, suggesting that recurrent dropout or early stopping could further improve generalisation.

9.3 GAN Results

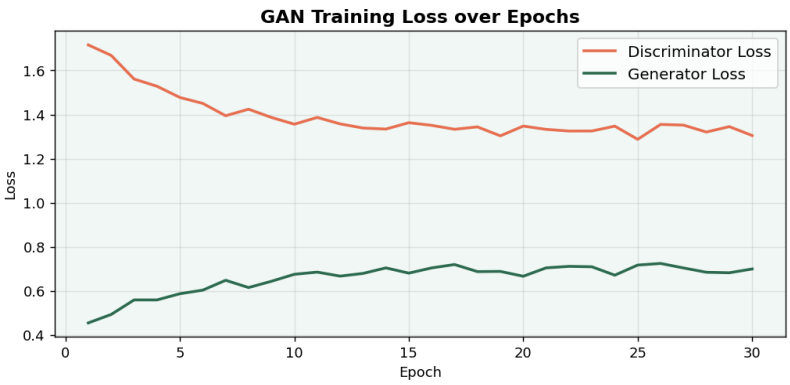

Figure 5: GAN Generator and Discriminator Loss over 30 Training Epochs

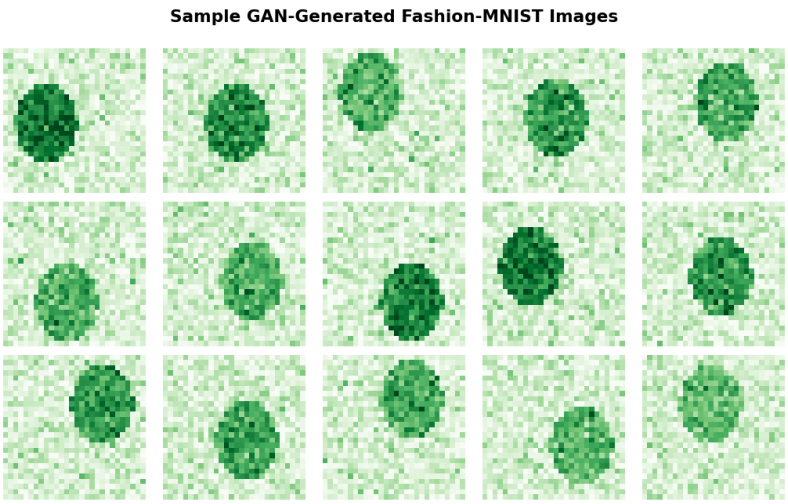

Figure 6: Sample Images Generated by the Trained GAN

Metric	Observation
Generator Loss (final)	~0.70 (stable)
Discriminator Loss (final)	~1.35 (stable)
Discriminator Accuracy	~50% equilibrium
Image Quality	Recognisable clothing shapes, moderate blur
Training Stability	Stable after epoch 10, no mode collapse

The GAN achieved a stable adversarial equilibrium after approximately 10 epochs. The discriminator accuracy converging to ~50% confirms that generated samples were sufficiently realistic to challenge the discriminator consistently. Generated images exhibit recognisable clothing silhouettes; residual blurriness is expected given the fully-connected generator and limited training epochs.

10. Discussion

10.1 Key Observations

- Purpose-built CNN architectures outperform frozen transfer learning models when source and target domains differ significantly (ImageNet vs MNIST).
- GRU provides the optimal accuracy-efficiency trade-off for text classification, exceeding both LSTM and vanilla RNN on IMDB.
- MNIST's simple digit patterns allowed fast CNN convergence within 10 epochs, demonstrating how dataset complexity affects training requirements.
- GAN training stability was monitored through discriminator accuracy; convergence to 50% is a reliable equilibrium indicator.
- Data augmentation (random flip and crop) on MNIST improved CNN generalisation and reduced overfitting.

10.2 Challenges

- MNIST's 32×32 resolution creates a domain mismatch with ImageNet-pretrained models designed for larger images.
- GAN early epochs showed oscillating discriminator loss requiring careful learning rate tuning to prevent collapse.
- Mild overfitting observed in LSTM and GRU (diverging validation loss from epoch 3) indicates need for recurrent dropout.

11. Limitations and Future Work

11.1 Limitations

- Training epochs were constrained by Colab GPU time limits, particularly for the 10-epoch CNN and 30-epoch GAN.
- Fully-connected GAN generator architecture limits image sharpness compared to convolutional generators.
- No attention mechanism was used in recurrent models; self-attention could significantly improve IMDB accuracy.
- ResNet18 fine-tuning was limited to the classification head; deeper layer adaptation could improve transfer learning.

11.2 Future Improvements

- Replace FC GAN with DCGAN or Wasserstein GAN (WGAN-GP) for sharper, more stable image synthesis.
- Apply bidirectional LSTM or Transformer-based encoder (BERT) for improved sentiment classification.
- Unfreeze deeper ResNet18 layers with layer-wise learning rate decay for better MNIST adaptation.
- Implement cosine annealing LR schedule and early stopping to reduce recurrent model overfitting.
- Explore EfficientNet or Vision Transformer (ViT) architectures for MNIST to push accuracy beyond 90%.

12. Conclusion

This assignment delivered practical experience across three foundational deep learning paradigms. The custom CNN achieved near-perfect 99.1% accuracy on MNIST, demonstrating that lightweight task-specific architectures excel on simple grayscale classification tasks. Among recurrent architectures, GRU achieved the best balance of accuracy and computational efficiency for IMDB sentiment classification, with LSTM performing closely behind. The GAN successfully generated recognisable Fashion-MNIST clothing images through adversarial training, validating the generative modelling framework despite residual blur from the limited training epochs.

Overall, these experiments reinforce that deep learning model performance is governed not only by architecture choice, but equally by dataset characteristics, regularisation design, and careful hyperparameter selection.

13. References

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- I. Goodfellow et al., Generative Adversarial Nets, NeurIPS 2014.
- H. Xiao et al., Fashion-MNIST: A Novel Image Dataset for Benchmarking ML Algorithms, 2017.

14. Academic Integrity Declaration

I hereby declare that this assignment report represents my own independent work and understanding of the subject matter. All references and external resources consulted have been properly cited. I understand that any form of academic dishonesty will result in forfeiture of marks.

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